Fiscal Year:	FY 2004	**Task Last Updated:**	FY 07/11/2005
PI Name:	Tryggvason, Gretar Ph.D.		
Project Title:	Splashing Droplets		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	gretar@wpi.edu	**Fax:**	FY 508-831-5680
PI Organization Type:	UNIVERSITY	**Phone:**	508-831-5759
Organization Name:	Worcester Polytechnic Institute		
PI Address 1:	Mechanical Engineering Department		
PI Address 2:	Higgins Labs 134, 100 Institute Road		
PI Web Page:			
City:	Worcester	**State:**	MA
Zip Code:	01609-2280	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	01/25/2001	**End Date:**	01/10/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Vander Wal , Randy	**Contact Phone:**	216-433-9065
Contact Email:	Randall.L.VanderWal@grc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kizito, John (NCMR c/o NASA-Glenn)		
Grant/Contract No.:	NAG3-2535		
Performance Goal No.:			
Performance Goal Text:			

The first year of the project will realize the hardware design and construction along with numerical simulation efforts for isothermal cases. During the second year survey measurements will begin in the 2.2 sec drop tower with aircraft flights as available. During the third and fourth years data collection will continue both in the 2.2 drop tower but primarily through aircraft flights. Predictive numerical simulations will be compared to experimental results as theory develops. Ultimately tests will be performed as suggested by theory for model evaluation. In all measurements, droplet fluid, size, velocity, and angle towards the surface and surface temperature will be systematically varied to demarcate boundaries between impingement regimes. It is expected that these boundaries will provide critical benchmark data to test theoretical simulations of the droplet impingement dynamics.
The main objective of the research is to mark the onset of different splashing behaviors and determine their temperature, pressure and angle dependence for impinging droplets of water, hydrocarbons and other fluids. A secondary objective will be the determination of the effect of liquid surface tension and viscosity upon the fingering evolution observed on

spreading droplet fronts and the transformation of these fingers into splashed products.

Task Description:

Current data on droplet breakup is scare for the sizes and velocities typical of practical applications such as in spray combustion processes and coating processes. The small spatial scales and rapid timescales representative of practical applications prevent detailed measurement of the internal fluid dynamics and liquid property gradients produced by droplets impinging upon surfaces. An improved understanding of drop breakup dynamics is needed to optimize the behavior of impinging droplets in these applications. Microgravity provides the environment in which to study large droplets with a shape undistorted by gravity. By trading off droplet size against velocity, large droplets with small velocities may be characterized by the same Weber and Reynolds numbers as small droplets at high velocities, characteristic of practical applications. The large size and slower velocities are expected to provide unparalleled spatial and temporal resolution of droplet surface impingement dynamics and thereby allow critical testing of theory. Finally a unique aspect of the work here will be to test equivalent dimensionless groups: for droplets of different size, adjustment of velocity, surface tension, and viscosity will be used to equalize Weber and Reynolds numbers to test for hidden property dependencies.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

The progress below extends over the duration of the grant.
Title: Droplets Splashing Upon Films of the Same Fluid of Various Depths

Conclusions

The impact of a droplet upon a thin film splashes where no splashing under the same conditions on a solid dry surface would. For impact conditions where the droplet would splash upon a dry solid surface, the film acts to enhance the splash dynamics. The direction, size and number of splashed products varies substantially, depending upon the presence and relative thickness of a surface film.

Our results show that for film thicknesses ranging from 0.1 = H* = 1, two splashing modes are observed, prompt and delayed -- associated with the advancing and receding crown respectively. For 1 = H* = 10, the liquid film acts to restrict prompt splashing and inhibits delayed splashing via crown formation and breakup for all but the highest energy impacts. In all cases, the occurrence of prompt splash is inconsequential in terms of fluid volume splashed relative to the initial droplet volume. Delayed splashing is of much greater consequence for splashed products. For H* ˜ 10, a central fluid jet is enabled by the greater liquid reservoir. Here the issue is not the number of ejected droplets but rather the existence of a jet. The prompt splash is similar to that with films of intermediate depth, while prompt and delayed splashing are minimal relative to that occurring on a thin film.

Both surface tension and viscosity affect the relative number and size of the ejected droplets. The number of ejected droplets during prompt and delayed splashing decreases while their mean size increases with increasing surface tension and viscosity. Comparison across the hydrocarbon series or water-based mixtures provides a range of viscosity values for fluids with similar surface tensions. Cross-comparison between decane and water or hexadecane and the 30% glycerol-water solution isolates the role of surface tension upon the splashing dynamics for fluids of similar viscosity. Surface tension appears to have little effect upon the crown splashing modes.

The role of viscosity is far more clear and consistent throughout the range of impact conditions studied here, 900 < Re < 14000 and 120 < We < 1400 for a 2 mm droplet. Increasing viscosity leads to a delay in both prompt and delayed crown splash, when they occur. More significantly, it correlates with a decreased number of splashed products and an increase in their size. Other effects such as crown wall thickness and uniformity were found to correlate with the fluid viscosity. The action of both fluid properties may be understood as acting to initiate and amplify Rayleigh-Taylor instability. The importance of determining different splashing modes is clearly based on many practical applications in addition to developing empirical correlations or models thereof. Such experimental guidance is critical to guiding theoretical development and providing definitive testing of its predictive capabilities.

In summary, upon a dry surface, the action of viscosity is to promote splashing by fostering a kinematic discontinuity in the advancing fluid front. In contrast, for droplets impinging upon a thin fluid film, the role of viscosity is reversed. The thin film itself interferes with the advancing fluid front to generate a kinematic discontinuity. Notably, such a discontinuity will be far larger than that between different locations in the fluid sheet derived solely from the droplet. Therein, a fluid with low viscosity and surface tension will experience the maximum deceleration at the advancing fluid front and hence splash prior to more viscous fluids. Interestingly, the role of surface tension appears to be consistent in both instances. High surface tension retards droplet release from either a spreading droplet upon a dry surface or from a crown rim as produced during impingement upon a liquid film. In other words, high surface tension inhibits splashing whether the receiving surface is dry or covered by a thin fluid film. When splashed product are produced there are fewer of them, they are larger, and their emission is delayed, in comparison to fluids with similar viscosity but lower surface tension.

Title: The Splash/Non-Splash Boundary Upon a Dry Surface and Thin Fluid Films.

Conclusions

Determining the splash/non-splash boundary is important for many practical applications. Painting, coating and cooling processes would each benefit from near-term empirical relations and subsequent models. Such demarcations can guide theoretical development by providing definitive testing of its predictive capabilities. Thus, empirical relations describing the boundary between splash and non-splash were given for drops impinging upon a dry solid surface and upon a thin fluid film covering a similar surface. Analytical simplification of the power laws describing the boundary between the splash and non-splash regions yields insight into the fundamental physical properties governing the splash and non-splash outcomes of the fluid droplets. Therein, splashing upon a dry surface is reasonably described by vCa > 0.85, reflecting the competing roles of surface tension and viscosity. Splashing upon a thin fluid film, as described by vv(rd/s) > 63, is governed by fluid density and surface tension, but is rather independent of viscosity. Finally, the data presented here suggests that a more direct dependence upon the surface tension and viscosity, given a better understanding of their interplay, would allow accurate description of the droplet surface impacts for more complicated situations involving non-Newtonian fluids, specifically those exhibiting viscoelastic behavior.

	Title: The Combined Influence of a Rough Surface and Thin Fluid Film Upon the Splashing Threshold and Splash Dynamics of a Droplet Impacting onto Them. Conclusions Both a thin fluid film and rough surface substantially alter the splashing threshold and subsequent splashing dynamics relative to a dry or smooth surface, respectively. A rough surface dramatically lowers the critical threshold for splash that is observed for a smooth surface. Secondly, a rough surface equalizes the drop impact dynamics due to fluid property variations. Differences in surface tension and viscosity become far less significant in either determining the splashing threshold or the subsequent dynamics which become very similar for all fluids. Differences in impact velocity merely alter the extension of the liquid filaments and hence the number of breakup products and to an extent the relative droplet sizes. The splashing dynamics upon the rough surface covered by a thin fluid film are a convolution of those which occur on a dry rough surface and a thin film upon a smooth surface. The consequences of each factor are evident in the type of crown formed and its subsequent breakup. As the velocity discontinuity along the fluid rim arises predominantly from the patternation of the rough surface, it is largely independent of drop impact energy or fluid properties. Hence, the splashing threshold is primarily determined by the surface features and secondarily by impact conditions and fluid properties. Thus, the main influence of fluid properties will be manifested in the breakup of the advancing fluid filaments. Acknowledgments This work was supported by a NASA NRA 98-HEDs-03 fluid physics award (RVW) administered through NASA cooperative agreement NAC3-975 with The National Center for Microgravity Research on Fluids and Combustion (NCMR) at the NASA-Glenn Research Center. The authors gratefully acknowledge Matthew Hall and Premal D. Patel for their many contributions to these experiments.
Bibliography Type:	Description: (Last Updated: 10/29/2019)
Articles in Peer-reviewed Journals	Tryggvason G, Bunner B, Esmaeeli A, Al-Rawahi N. "Computations of Multiphase Flows." Advances in applied mechanics. 2003 December 39;81-120. https://doi.org/10.1016/S0065-2156(02)39002-1 , Dec-2003
Articles in Peer-reviewed Journals	Esmaeeli A, Tryggvason G. "Computations of Explosive Boiling in Microgravity." Journal of Scientific Computing. 2003 December, Volume 19, Issue 1–3, pp 163–182 , Dec-2003
Presentation	Randy L. Vander Wal, John P. Kizito, Gordon M. Berger and Steven D. Mozes "Droplet-Surface Impingement Dynamics for Intelligent Spray Design and Operation" 2004 Painting Technology Workshop, Lexington, KY Sep-2004
Presentation	Randy L. Vander Wal, John P. Kizito, Gordon M. Berger and Steven D. Mozes "Droplet-Surface Impingement Dynamics" Strategic Research to Enable NASA's Exploration Missions Conference, Cleveland OH NASA/TM-2004-213114. Jun-2004
Presentation	Randy L. Vander Wal, John P. Kizito, Gordon M. Berger and Steven D. Mozes "Droplet-Film Splash Dynamics" Stability and Nonlinear Hydrodynamics Session at The 2003 AIChE Annual Fall Meeting, San Francisco, CA Nov-2004